

Quantum informatics

Quantum communication and computing

Aurél Gábris



Faculty of Nuclear Sciences and Physical Engineering
Czech Technical University in Prague



Wigner Research Centre for Physics



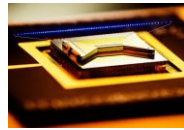
<https://www.qworld.net>



<https://qczech.org>

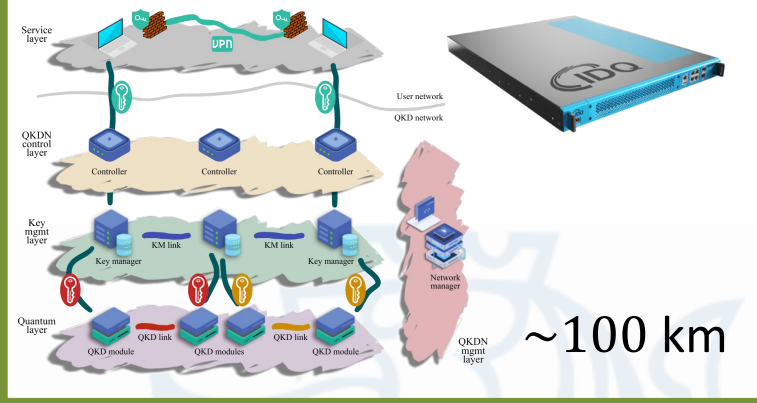
Quantum informatics at glance

Quantum Computing



~500 qubits

Quantum Cryptography



Quantum Technologies

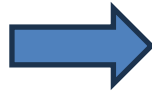
Fabrication, material science, electronics, cooling,
vacuum tech, photonics (lasers, detectors)

“Quantum 2.0”

quantum physics:
discoveries, modelling

What for Quantum Computing?

Totally new architecture



- New algorithms
- New complexity classes ($P \rightarrow BQP$)

More efficient

- exponentially (future)
- quadratically (present)

Factorization

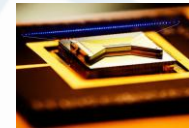
- Threat to RSA, DH
- In 10-30 years



Peter Shor

“Harvest now decrypt later”

Quantum Computing



~500 qubits

What for Quantum Cryptography?

Quantum computer

- Break RSA in 10-30 years
- Harvest now decrypt later



Data shelf-life > 30 years

Imminent danger

Mitigate threat

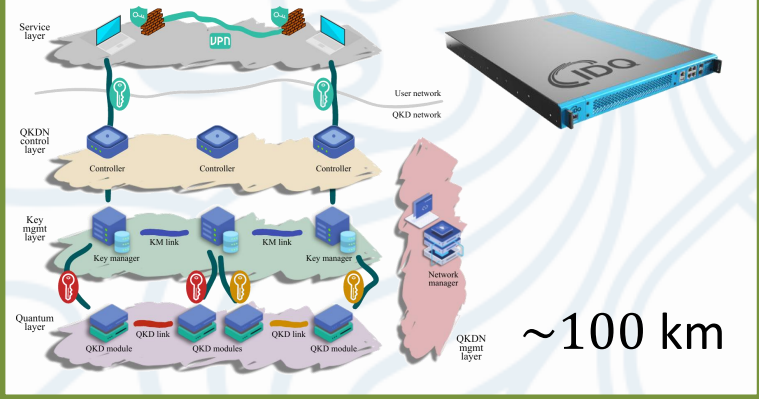
“Post-quantum cryptography”

- new crypto algorithms
- lattice-based, etc...

Quantum cryptography

- physically guaranteed security

Quantum Cryptography



Today: Quantum cryptography

But before...

I. Information is physical

- *Lessons from the history of computers*
- *Semiconductor technology (Quantum 1.0), Moore's law*

II. Quantum 2.0

- *The quantum bit*
- *Superposition, measurement*

III. Quantum cryptography

- *Quantum key distribution: BB84*
- *QKD in action and challenges*

I. Information is physical

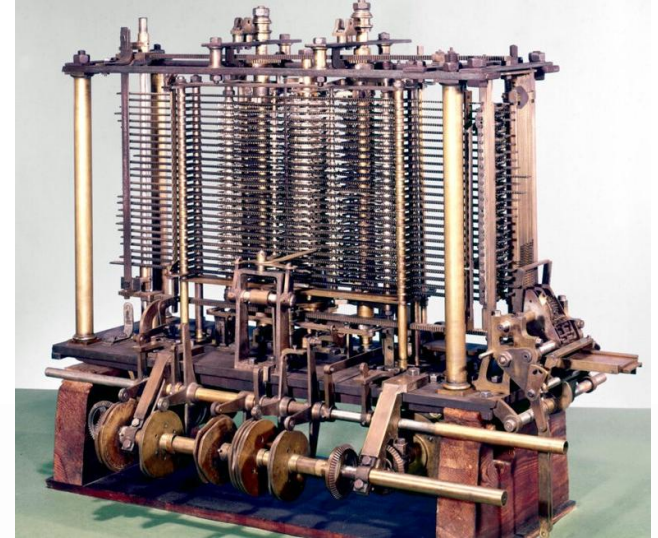


Mechanical computers

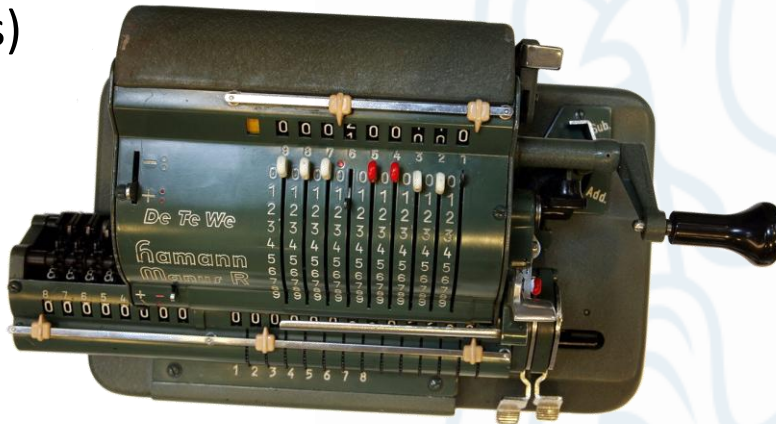
Babbage's differential engine (1822)

<https://www.youtube.com/watch?v=BlbQsKpg3Ak&t=785s>

5 tons (8000 pieces)

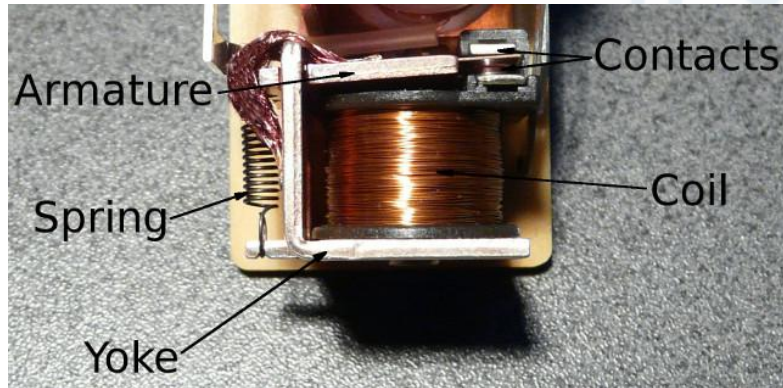
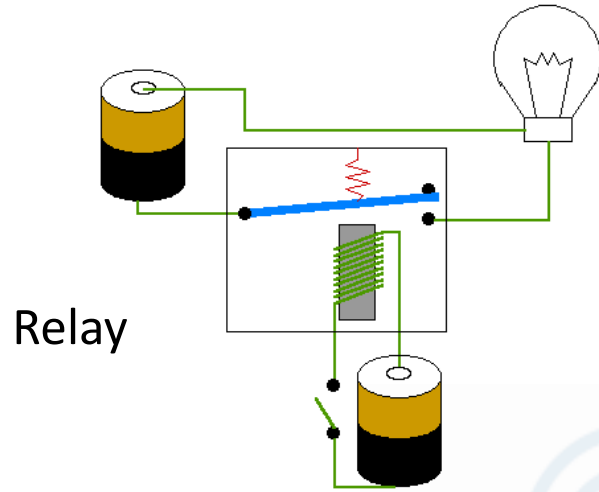


Hamann Manus R
(1950's)



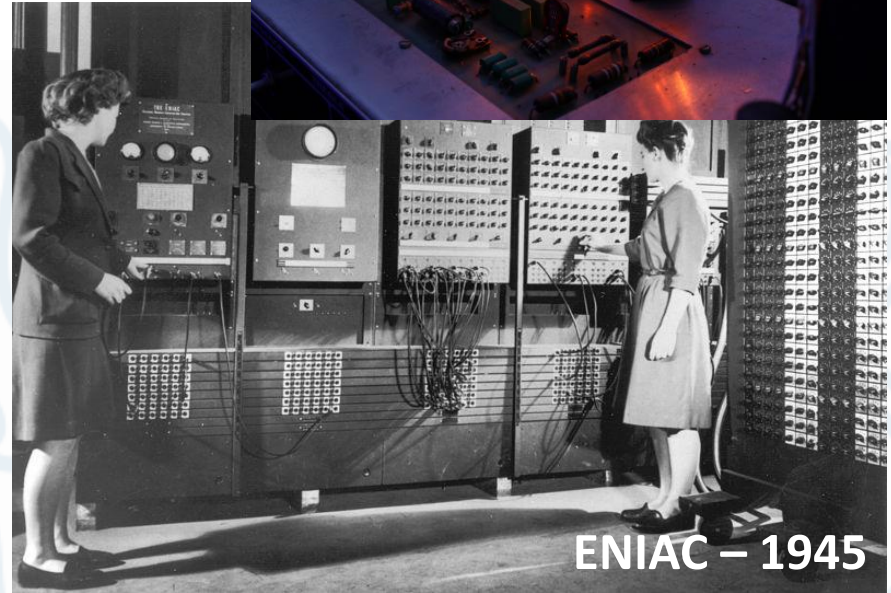
decimal representation

Electro-mechanical



Electrical

Vacuum/electron
tube



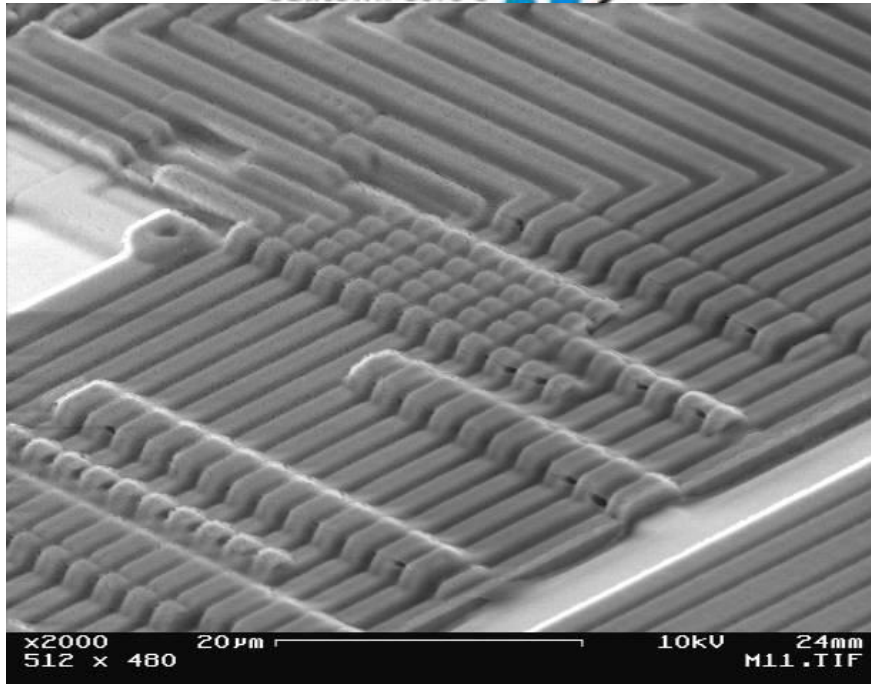
Semiconductor technology

“quantum 1.0”

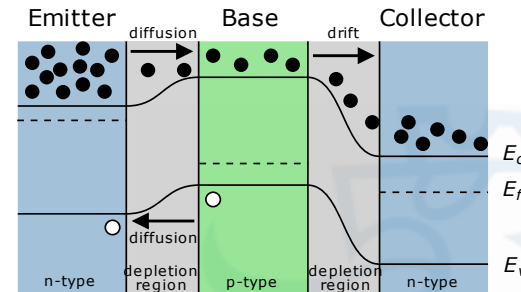
Moore's Law

Gulftown Core 6

Ivy Bridge



Semi-classical approximation

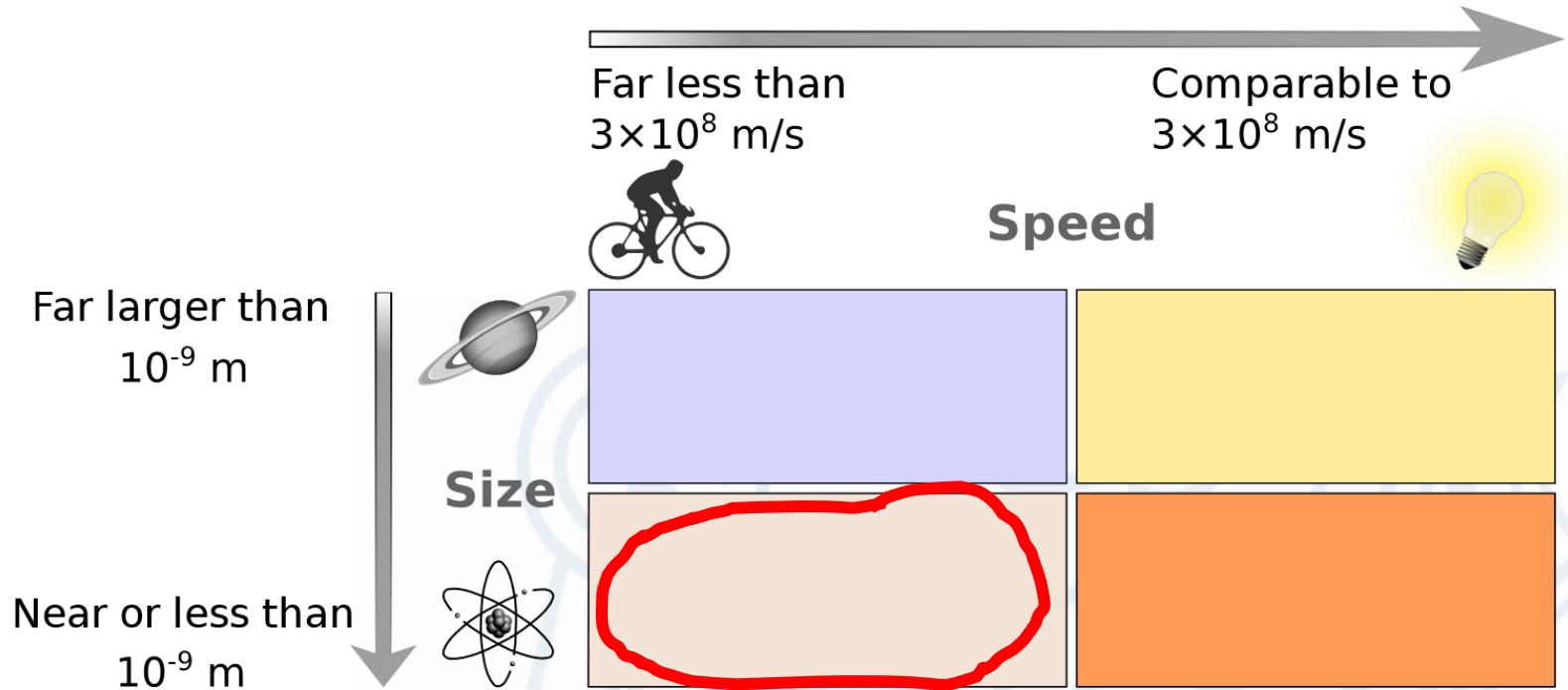


Transistors are smaller and smaller
They become too quantum!
(approximation breaks down)

II. Quantum 2.0

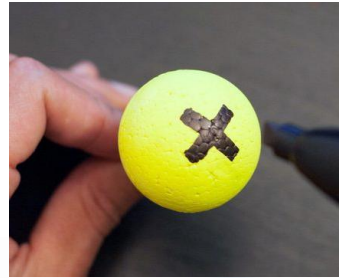
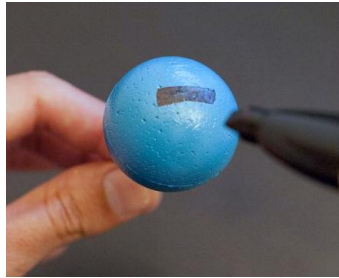


What is “quantum?”

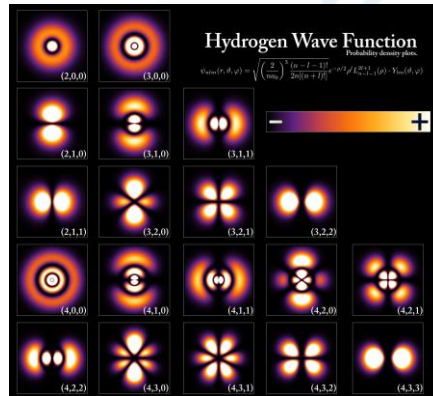


Quantum theory: the theory of the small

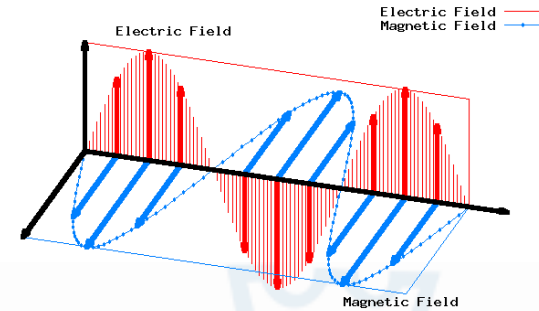
- Elementary particles



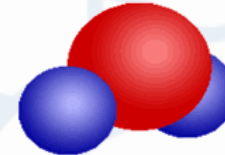
- Atoms/ions



- Electromagnetic field

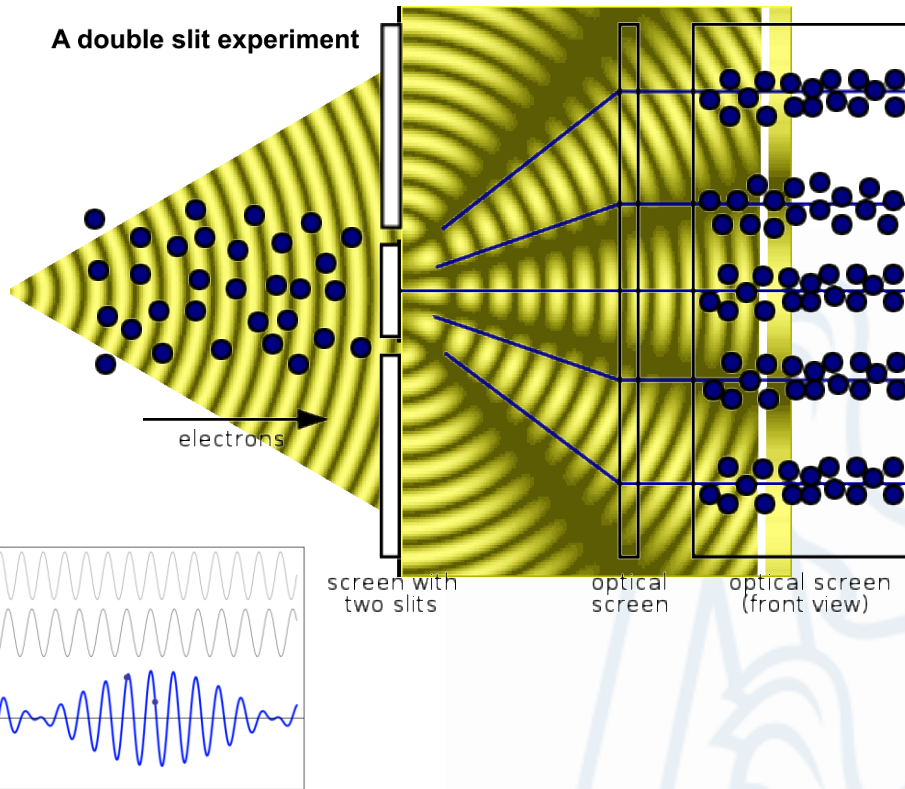


- Molecules



Superposition

- Wave – particle duality



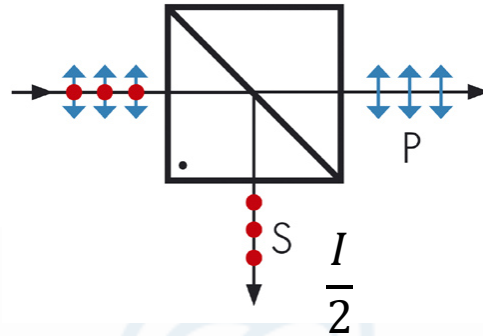
superposition



Splitting the photon

Light waves

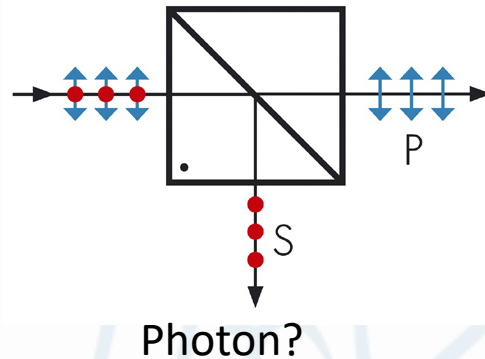
Intensity: I



$$\frac{I}{2}$$

Beam split in half

Single photon



Photon?

Probability: $\frac{1}{2}$

Probability vs. amplitude

With probabilities

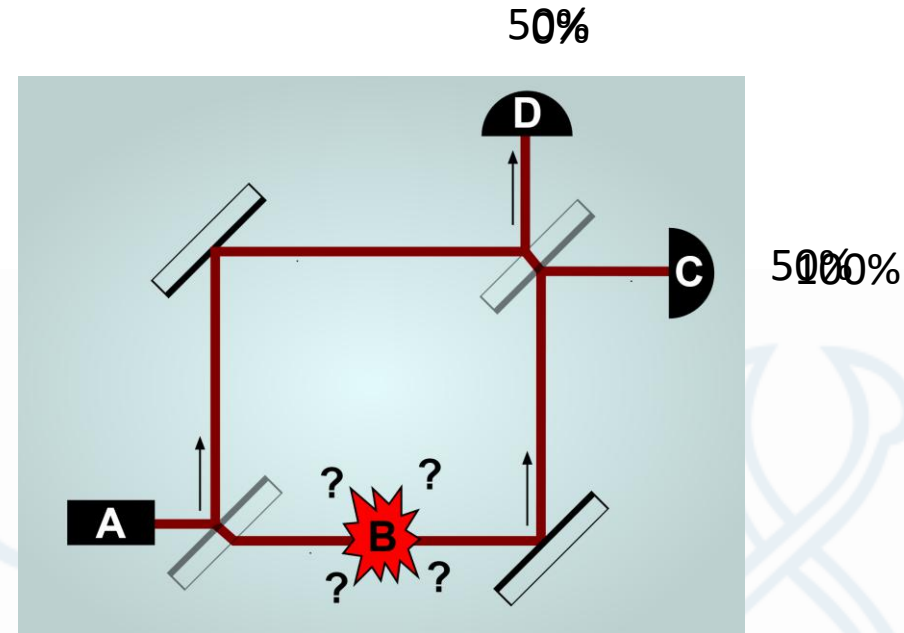
- $\mathcal{P}(0) = (1,0)$
- $\mathcal{P}(1) = (1/2, 1/2)$
- $\mathcal{P}(2) = (1/4 + 1/4, 1/4 + 1/4) = (1/2, 1/2)$

$$\frac{1}{2} + \frac{1}{2} = 1$$

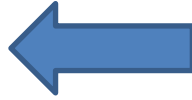
With amplitudes

- $\psi(0) = (1,0)$
- $\psi(1) = (1/\sqrt{2}, i/\sqrt{2})$
- $\psi(2) = \left(\frac{i}{2} - \frac{-i}{2}, \frac{i}{2} + \frac{-i}{2}\right) = (1,0)$

$$\left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{i}{\sqrt{2}}\right|^2 = 1$$

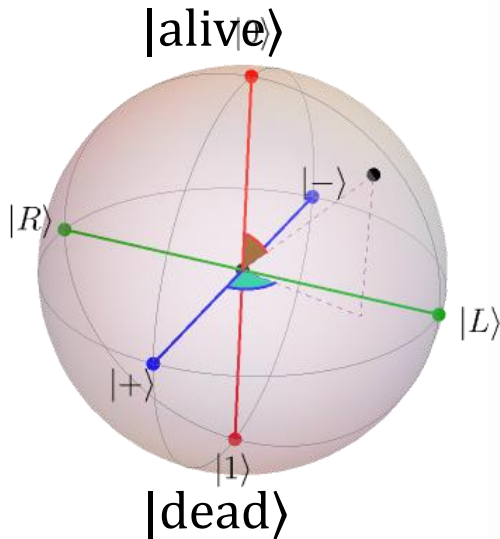


Measurement and “collapse”

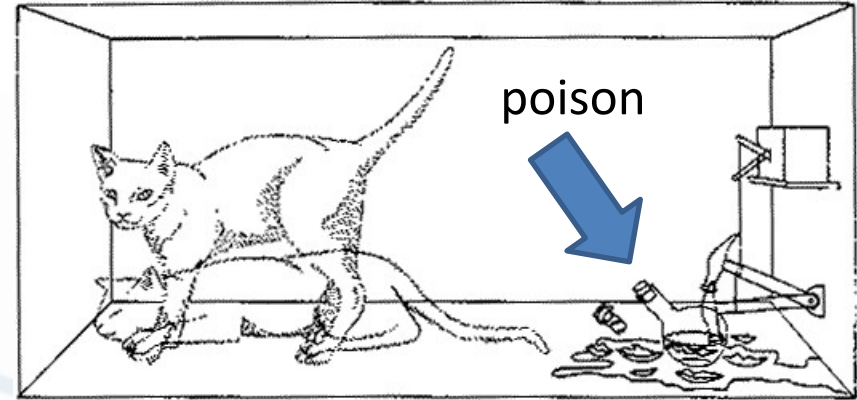


Schrödinger's cat

superposition



dead
AND
alive



after opening box:

- only “dead” or “alive”
- no superposition

Probabilistic bit: “pbit”

Bit: states

- “zero” or “one”
- “on” or “off”
- $V < V_0$ or $V > V_0$

0

we know the “state”

1

Probabilistic bit

we don’t know the “state”

- coin after flipping
- we can tell how likely it is to be “heads” or “tails”, resp. 0 or 1
- probability of 0: $\mathcal{P}(0) = p_0$
- probability of 1: $\mathcal{P}(1) = p_1$

Probabilistic vector

One bit:

$$[p] := \begin{pmatrix} p_0 \\ p_1 \end{pmatrix}$$
$$\sum_{j=0}^1 p_j = 1$$

probability of 0: $\mathcal{P}(0) = p_0$

probability of 1: $\mathcal{P}(1) = p_1$

total probability is 1

$$\mathcal{P}(j) = p_j \geq 0, \quad (\forall j = 0, 1)$$

Basis in the vector space

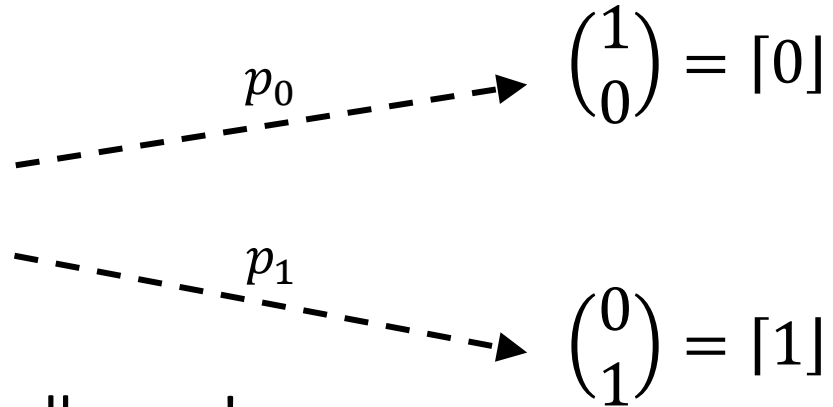
$$[\mathbf{0}] := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$[\mathbf{1}] := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Measurement of pbit

State:

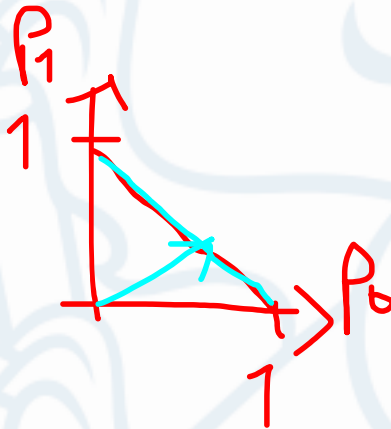
$$[p] := \begin{pmatrix} p_0 \\ p_1 \end{pmatrix}$$



After "read out" pbit state collapses!

Final state: $[j]$

- Projection operator
- Re-normalization



$$(1,0) \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} = p_0$$

$$(0,1) \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} = p_1$$

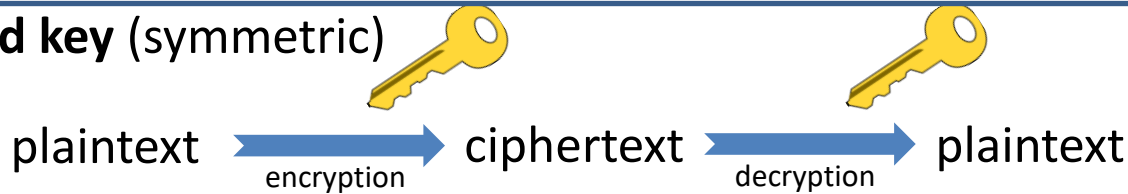
III. Quantum cryptography



Cryptography in a nutshell

Goal: send message so that only the recipient can read it

Shared key (symmetric)



security:



kept secret

weakness:

distribute



securely?

should not be reused



Public key (asymmetric)



RSA

<https://>

security:

computational
hardness

threat:

newly invented algorithms

One-time pad (Vernam cipher)

$\text{length}(\text{key}) = \text{length}(\text{plaintext})$

Plaintext	Key
H - 01001000	01010000
E - 01010000	01001100
L - 01001100	01010101
L - 01001100	01010100
O - 01001111	01001111

XOR / modulo 2 addition

$$(A \oplus B) \oplus B = A$$

encryption

decryption

apply twice → recover original

INPUT		OUTPUT
A	B	$A \oplus B$
-	-	-
-	+	+
+	-	+
+	+	-

plaintext	01001000	01010000	01001100	01001100	01001111
key	01010000	01001100	01010101	01010100	01001111
ciphertext	00011000	00011100	00011001	00011000	00000000

UNBREAKABLE
as long as key stays secret

how to distribute
such a long key?



Quantum key distribution: BB84

Sending bits in two basis: $\{|0\rangle, |1\rangle\}$ and $\{|+\rangle, |-\rangle\}$

$$|\pm\rangle := (|0\rangle \pm |1\rangle)/\sqrt{2}$$

not orthogonal: $|\langle\pm|0\rangle|^2 = 1/2$

Random bit strings: Alice: a, b
(length k) Bob: b'

(a_j, b_j)

Alice

$|\psi_{a_j b_j}\rangle$

b'_j

Bob

a'_j

$$b'_j = b_j: a'_j = a_j$$

$$b'_j \neq b_j: a'_j = \text{random}$$

$$|\psi_{00}\rangle = |0\rangle$$

$$|\psi_{10}\rangle = |1\rangle$$

$$|\psi_{01}\rangle = |+\rangle$$

$$|\psi_{11}\rangle = |-\rangle$$

$$b'_j = 0: \{|0\rangle, |1\rangle\}$$

$$b'_j = 1: \{|+\rangle, |-\rangle\}$$

Quantum bit (qubit)

usual bit:

0

1

0: $V > V_0$

1: $V < V_0$

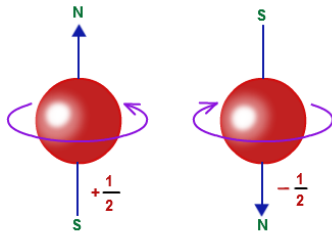
qubit:

$|0\rangle$

$|1\rangle$



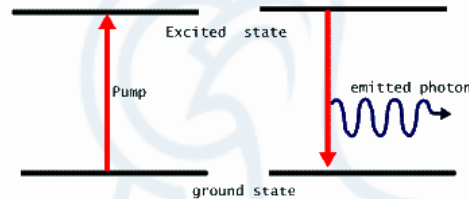
Electron spin



$|\uparrow\rangle$

$|\downarrow\rangle$

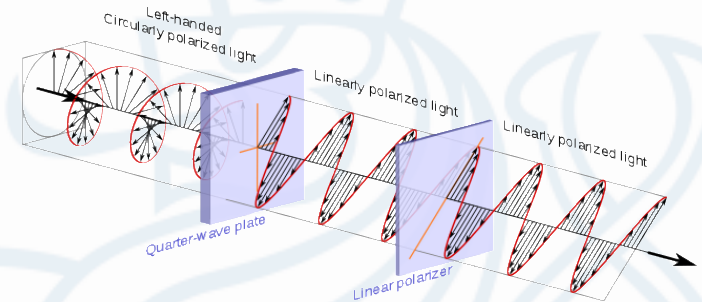
electronic structure of atom



$|e\rangle$

$|g\rangle$

polarization of photon



$|\updownarrow\rangle$

$|\leftrightarrow\rangle$

Quantum bit

Amplitudes

$$\psi = (\alpha, \beta)$$

α, β complex numbers

$$|\alpha|^2 + |\beta|^2 = 1$$

Basis states in $\mathbb{C}^2$

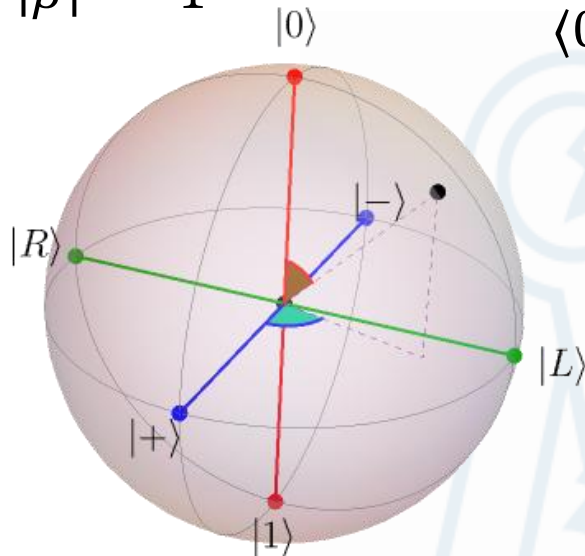
$$|0\rangle = (1, 0)$$

$$|1\rangle = (0, 1)$$

$$\langle 0|1\rangle = 0$$

Another basis in $\mathbb{C}^2$

$$\begin{aligned} |\pm\rangle &= \frac{1}{\sqrt{2}} (1, \pm 1) \\ &= \frac{1}{\sqrt{2}} (|0\rangle \pm |1\rangle) \end{aligned}$$

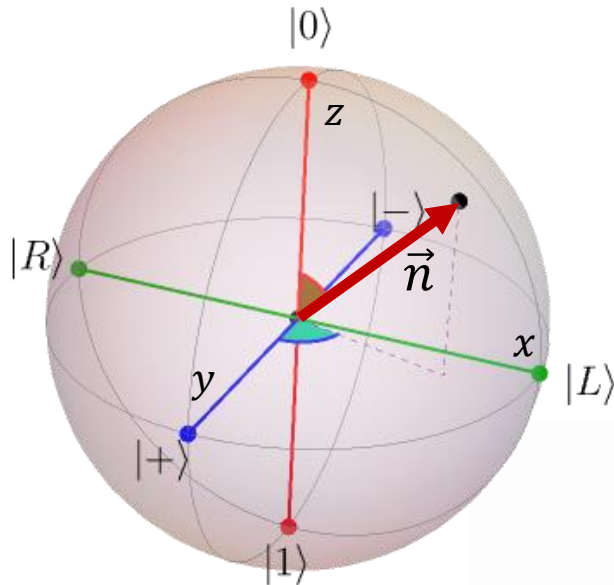


Dirac notation

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

<https://javafxpert.github.io/grok-bloch/>

State of qubit: Bloch sphere



$$\begin{aligned} |\psi\rangle &= \alpha_0|0\rangle + \alpha_1|1\rangle \\ &= \cos \theta/2 |0\rangle + e^{i\varphi} \sin \theta/2 |1\rangle \end{aligned}$$

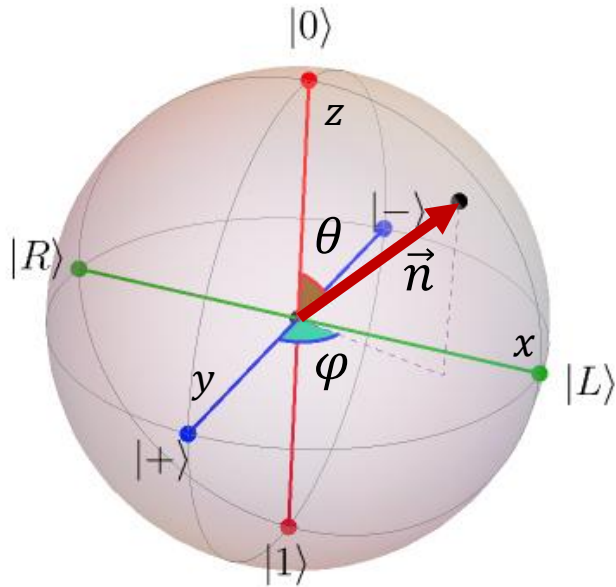
point on the surface of sphere

Spherical coordinates: $\vec{n} = (r, \theta, \varphi)$
 $\theta \in [0, \pi], \varphi \in [0, 2\pi], r = 1$

$$\begin{aligned} |\pm\rangle &= \frac{|0\rangle \pm |1\rangle}{\sqrt{2}} \\ |R/L\rangle &= \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}} \end{aligned}$$

We have used: $|\alpha|^2 + |\beta|^2 = 1$ “normalization”
 $\alpha \in \mathbb{R}$ “global phase”

Operations



$$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$$

point on the surface of sphere

Operations: unitary

$$U^\dagger U = \mathbb{I}$$

Rotation of Bloch vector

“computational basis”

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$U = \begin{pmatrix} u & v \\ -v^* & u^* \end{pmatrix}$$

$$|u|^2 + |v|^2 = 1$$

$SU(2)$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Measurement

Two orthogonal states: $\langle + | - \rangle = 0$

$$|\pm\rangle = \frac{1}{\sqrt{2}} (|0\rangle \pm |1\rangle)$$

x axis on the Bloch sphere

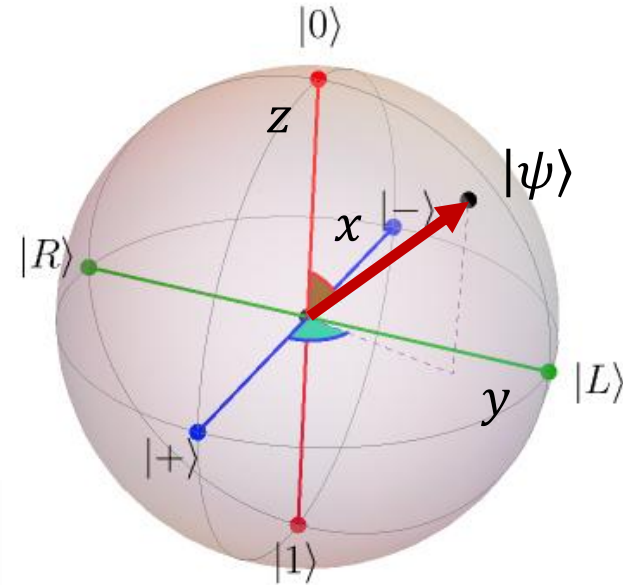
Outcome probability:

$$|\langle 0 | + \rangle|^2 = \frac{1}{2}$$

Δ : enclosed angle

Everything can be understood in terms of rotations

- Light polarization: Poincaré sphere
- Spin: physical space



Two qubits

Single qubit: $|\psi\rangle \in \mathcal{H}_2$

Two qubits: $|\psi\rangle \in \mathcal{H}_2 \otimes \mathcal{H}_2$

n qubits: $|\psi\rangle \in \mathcal{H}_2^{\otimes n}$



Kronecker product

The basis by elementary tensor products:

$$\begin{aligned} |00\rangle &= |0\rangle \otimes |0\rangle = |0\rangle|0\rangle \\ |01\rangle &= |0\rangle \otimes |1\rangle = |0\rangle|1\rangle \\ |10\rangle &= |1\rangle \otimes |0\rangle = |1\rangle|0\rangle \\ |11\rangle &= |1\rangle \otimes |1\rangle = |1\rangle|1\rangle \end{aligned}$$

“binary” notation
(0,1,2,3)

shorthand

Copying quantum states?

Qubit in state

- $|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$

If we knew α_0 and α_1 **then** we could prepare, say, from $|0\rangle$.

By measurement we cannot learn these parameters!

projection to $|0\rangle$ or $|1\rangle$

Direct copy by quantum circuit?

Is there some unitary operation U , such that:

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle?$$

No! “no cloning theorem”

No-cloning theorem

Cloning should work for any states $|\psi\rangle$ and $|\phi\rangle$

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle, \quad U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle$$

Initial scalar product

$$\langle\phi|\langle 0|\psi\rangle|0\rangle = \langle\phi|\psi\rangle\langle 0|0\rangle = \langle\phi|\psi\rangle$$

Final scalar product

$$\langle\phi|\langle 0|U^\dagger U|\psi\rangle|0\rangle = \langle\phi|\psi\rangle\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2$$

However, by unitarity ($U^\dagger U = \mathbb{I}$)

$$\langle\phi|\langle 0|U^\dagger U|\psi\rangle|0\rangle = \langle\phi|\langle 0|\psi\rangle|0\rangle = \langle\phi|\psi\rangle$$



$$\langle\phi|\psi\rangle=1 \quad |\psi\rangle = |\phi\rangle$$

or

$$\langle\phi|\psi\rangle=0$$

Does not work for arbitrary $|\psi\rangle$

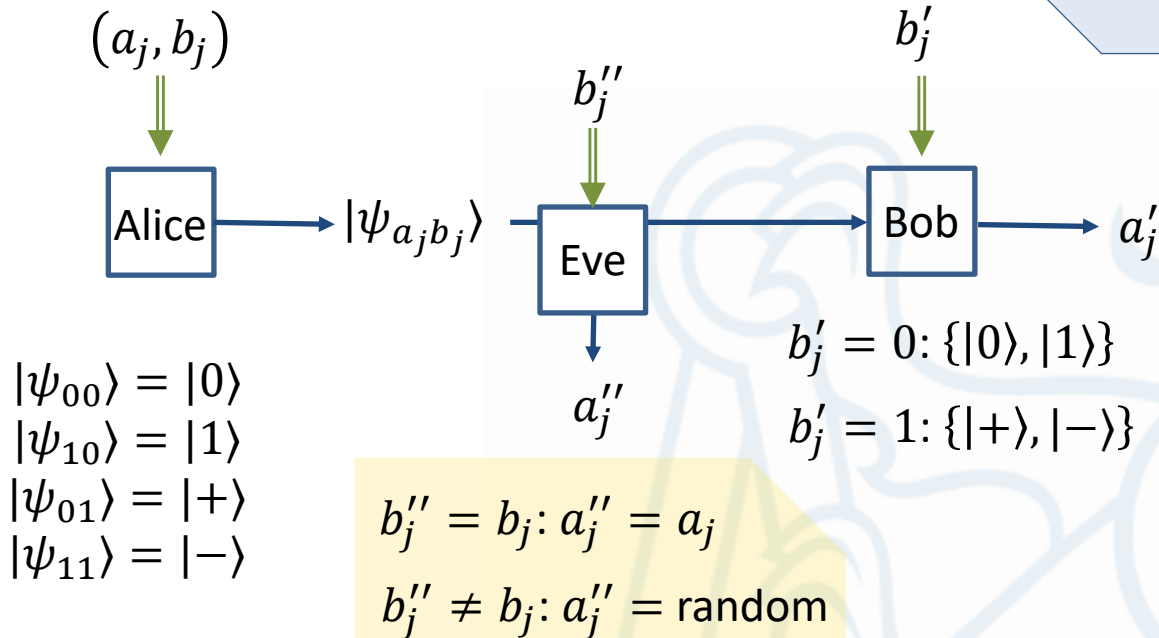
Quantum key distribution: BB84

Sending bits in two basis: $\{|0\rangle, |1\rangle\}$ and $\{|+\rangle, |-\rangle\}$

$$|\pm\rangle := (|0\rangle \pm |1\rangle)/\sqrt{2}$$

not orthogonal: $|\langle\pm|0\rangle|^2 = 1/2$

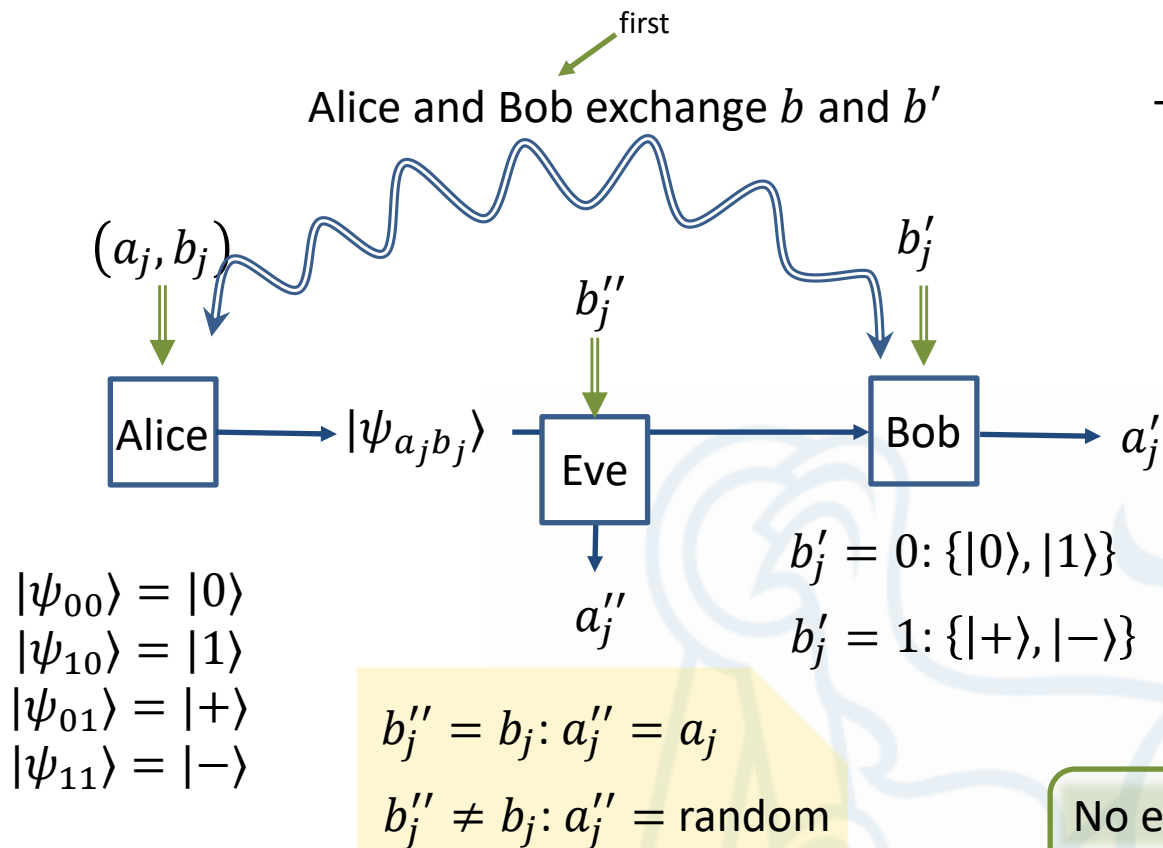
Random bit strings: Alice: a, b
(length k) Bob: b'



$b'_j = b_j: a'_j = a_j$
 $b'_j \neq b_j: a'_j = \text{random}$

$b''_j \neq b_j:$
 $b'_j = b_j: a'_j = \text{random}$
 $b'_j \neq b_j: a'_j = \text{random}$

BB84: Announcement phase



They keep a_j and a'_j only for j where $b'_j = b_j$

b_j	b'_j	a_j	a'_j
1	1	✓	✓
0	1	✗	✗
1	1	✓	✓
0	0	✓	✓
1	0	✗	✗

No errors, no eavesdropping
 $a = a'$

BB84: Detecting eavesdropping

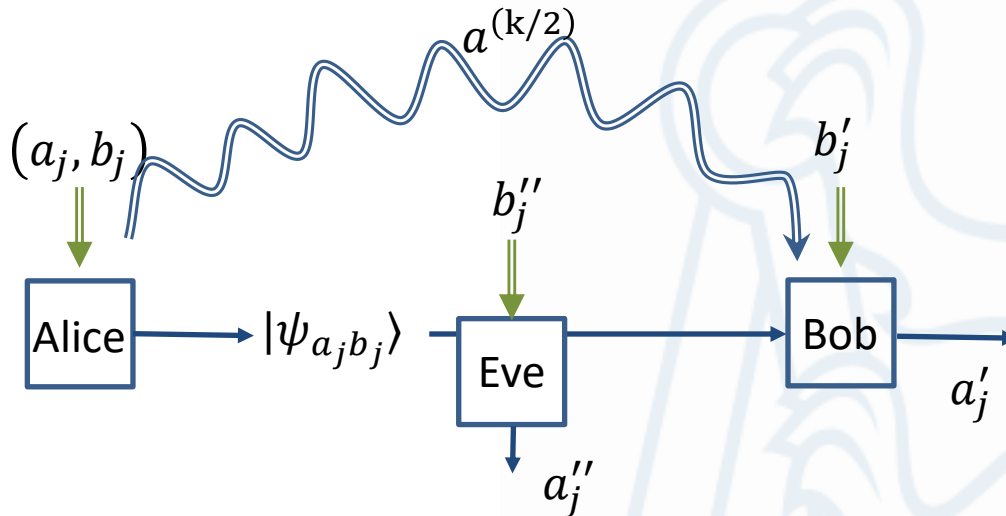
Error introduced by Eve: at most 25%

- ✓ $\frac{1}{2}$ time Eve will get b_j right
- $\frac{1}{2}$ time will miss it
 - ✓ $\frac{1}{2}$ time Bob will get a_j right
 - $\frac{1}{2}$ time will get it wrong

Estimating error rate: η

Alice announces half of a
($k/2$ elements at random)

Bob checks against his a' :
counts incorrect positions

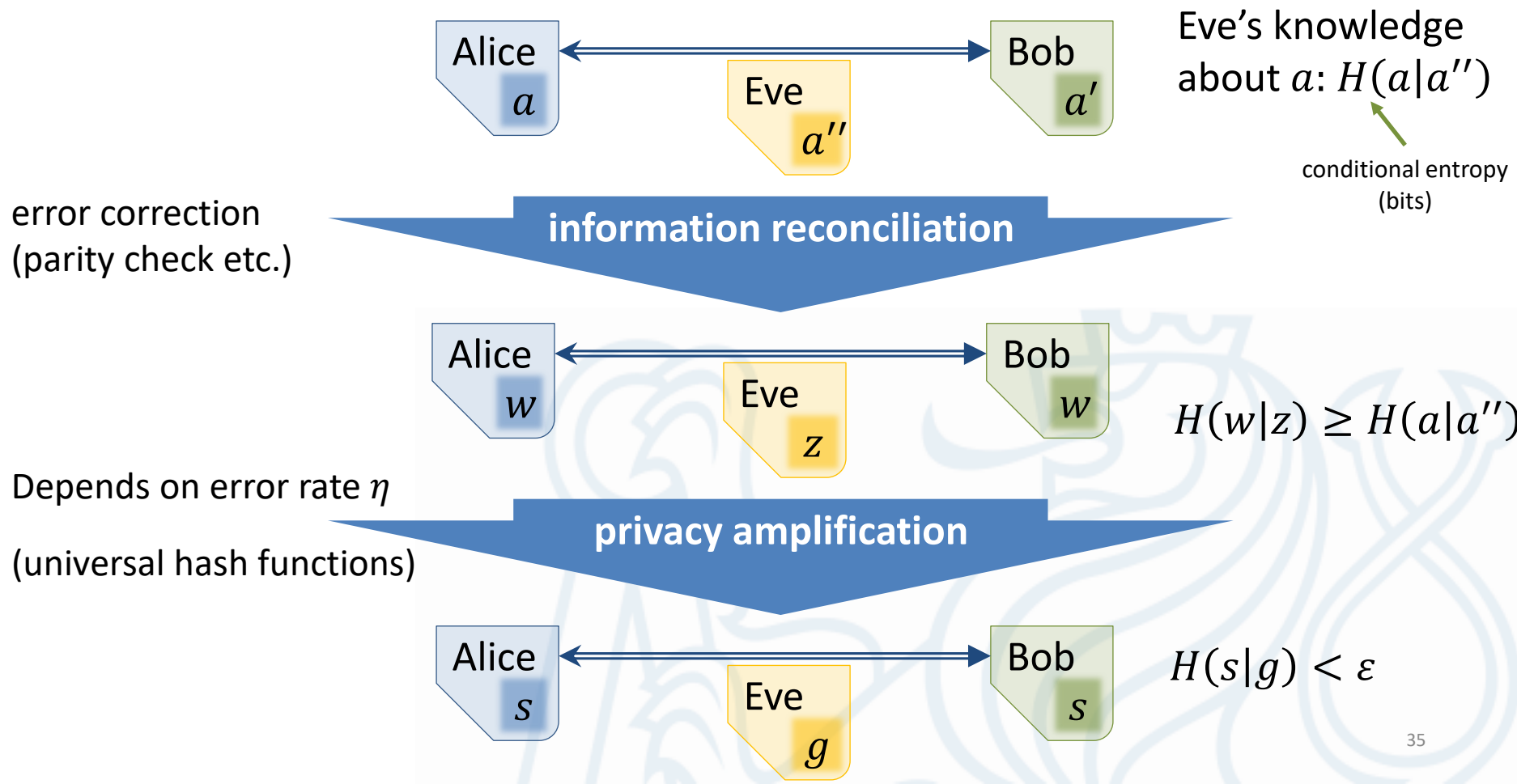


$\eta \geq 25\%$

suspend communication

No errors, no eavesdropping
 $a = a'$

Mitigating eavesdropping



Global implementation EuroQCI



China

Shaping Europe's digital future

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The European Quantum Communication Infrastructure (EuroQCI) Initiative

The EuroQCI will be a secure quantum communication infrastructure spanning the whole EU, including its overseas territories.

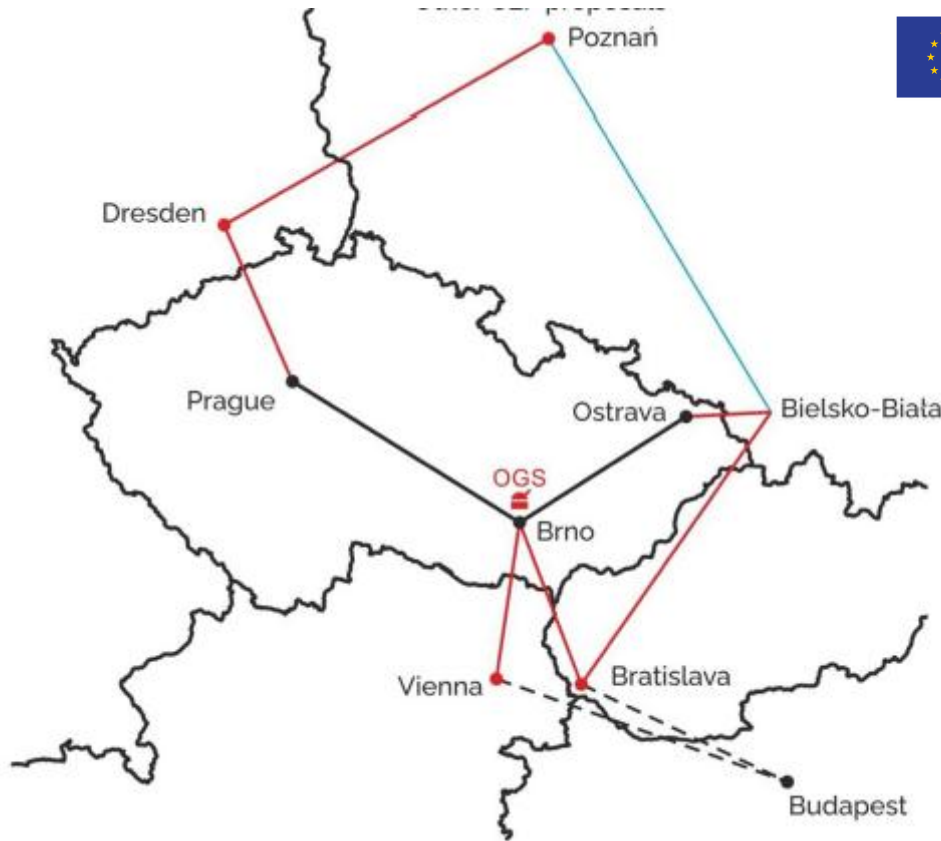


length: 2000km
nodes: >30

"trusted node"

National QKD infrastructure

CZQCI



Co-funded by
the European Union



NÁRODNÍ
PLÁN OBNOVY

NÚKIB 

CyberSecurityHub^{CZ}



MASARYKOVA
UNIVERZITA



Palacký University
Olomouc

VSB TECHNICAL
UNIVERSITY
OF OSTRAVA

IT4INNOVATIONS
NATIONAL SUPERCOMPUTING
CENTER

VYSOKÉ UČENÍ
TECHNICKÉ
V BRNĚ



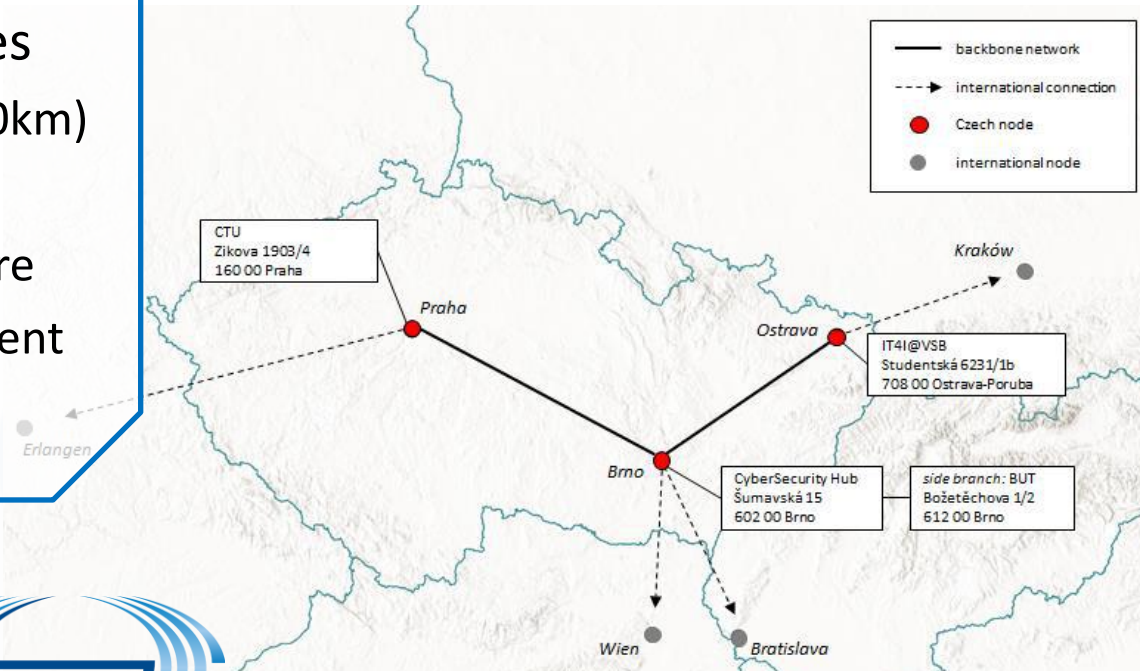
ISI
CAS
Institute of Scientific
Instruments
The Czech Academy
of Sciences

cesnet

National QKD infrastructure

CZQCI

- QKD technology features
 - Limited distance (80-100km)
 - Dedicated optical fibre
 - Trusted node architecture
 - Quantum key management system (KMS)

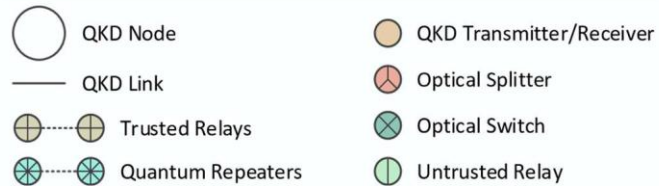
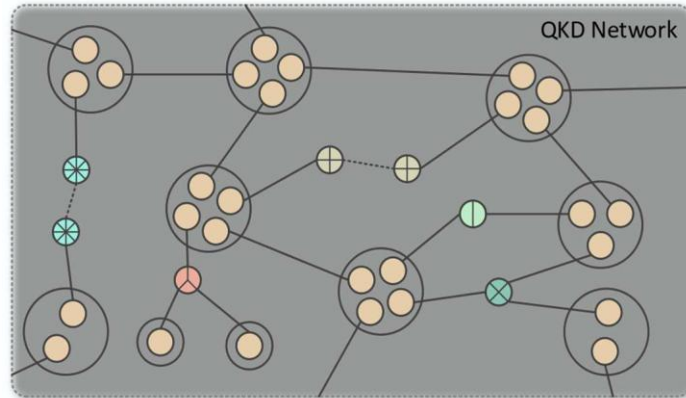


Interface:

- REST API
- ETSI GS QKD 014

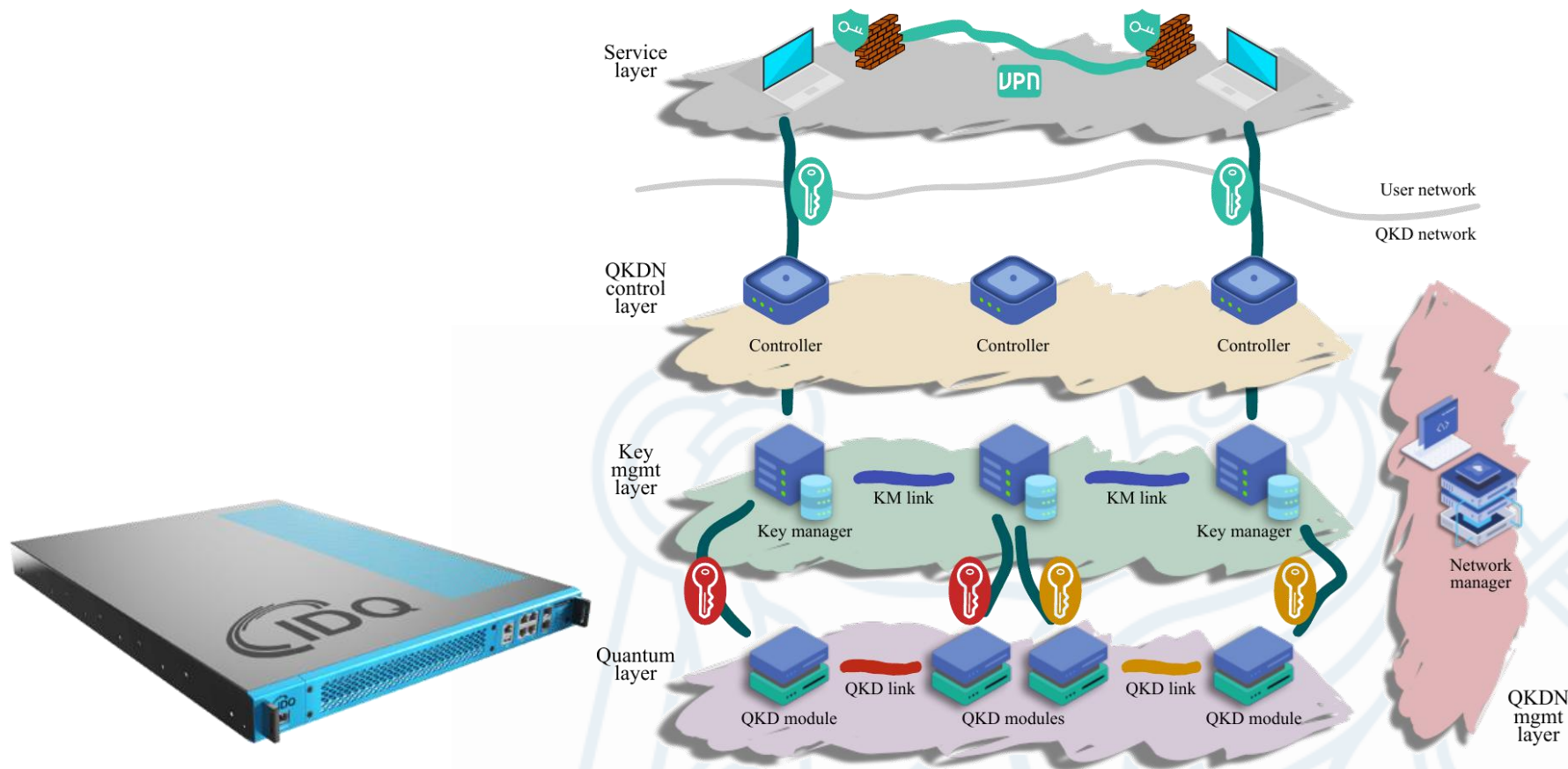


Implementation

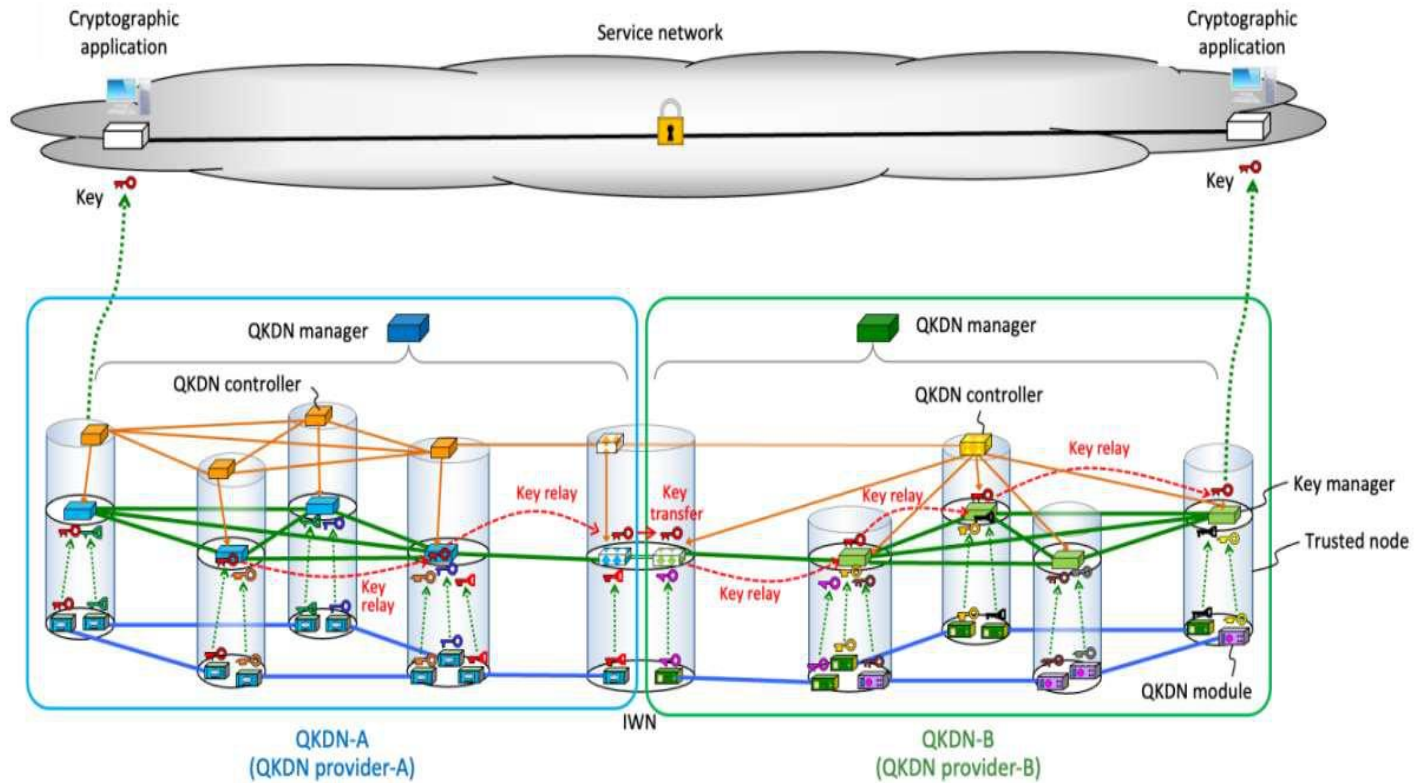


	Optical switching	Trusted relay-based	Untrusted relay-based	Quantum repeater
Achievable distance	Relatively low	Relatively high	Relatively low	High
Scalability	Relatively low	High	Relatively low	High
Applicability	Controller	Controller	Controller	Wide
Security	High	Relatively high	High	High
Maturity	High	High	Relatively low	Low
Field trial	QKD module	QKD link	QKD module	Unavailable

Implementation



Implementation



Beyond the trusted-node architecture

Decoherence

$$|\psi_{01}\rangle = |+\rangle = (|0\rangle + |1\rangle)/2$$



$$\rho = \begin{cases} |0\rangle \text{ with probability } 1/2 \\ |1\rangle \text{ with probability } 1/2 \end{cases}$$

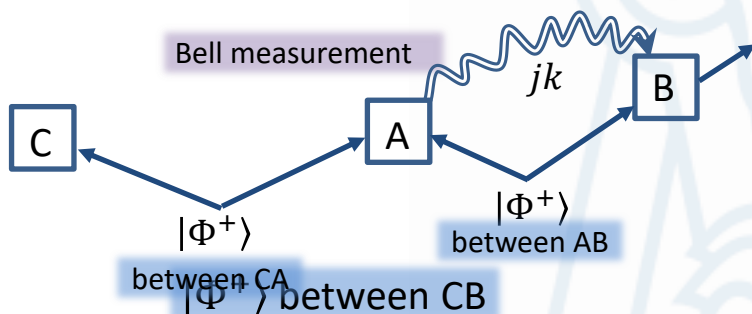
density
operator

probabilities	σ_z	σ_x
0 (+1)	$1/2$	$1/2$
1 (-1)	$1/2$	$1/2$

← measurement basis

outcome →

Entanglement swapping



Entanglement distillation

entanglement degrades (decoherence)
many “bad” entangled states



fewer $|\Phi^+\rangle$

Next week

How to program quantum computers?

- Jupyter notebook
- qiskit



Qiskit



Jupyter

Create small quantum circuits

- See the coding principles
- experience quantum features

Will be sending links to notebooks before the lecture

All else you need to know is



pythonTM